CURRICULUM VITAE Michael Neunteufel

Contact information

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Born on March 1, 1994. Austrian citizenship

Personal profile

I published 26 peer-reviewed publications, collaborated with 25+ researchers, reviewed 25+ papers, and gave 30+ talks in wide-range applications of finite element methods including nonlinear continuum mechanics, shells, fluid-structure interaction, shape optimization, and curvature approximation. Since 9 years I have been a developer of the FEM software NGSolve and I am the maintainer of NGSDiffGeo.

Keywords: finite element method, elasticity, shells, differential geometry, shape optimization, fluid-structure interaction, numerical relativity

Education

2017-2021 TU Wien (Austria), **Ph.D. in Technical Mathematics (with highest national distinction)**

Thesis title: "Mixed Finite Element Methods For Nonlinear Continuum Mechanics And Shells"
Thesis supervisor: Prof. Joachim Schöberl
Referees: Prof. Antonio Gil (Swansea University), Prof. Max Wardetzky (Univ-Göttingen), Prof. Astrid Pechstein (JKU Linz)

2016-2017 TU Wien (Austria), **M.Sc. in Technical Mathematics (with distinction)**

Thesis title: "Advanced Numerical Methods for Fluid Structure Interaction"
Thesis supervisor: Prof. Joachim Schöberl

2013-2016 TU Wien (Austria), **B.Sc. in Technical Mathematics (with distinction)**

Thesis title: "TRNNM - Gemischte Finite Elemente für nichtlineare Schalenmodelle"
Thesis supervisor: Prof. Joachim Schöberl

Employment

Dec 2025-present	TU Wien (Austria) **Erwin Schrödinger Fellow** (postdoctoral researcher)
Sep 2024-Nov 2025	Portland State University (OR, USA) **Erwin Schrödinger Fellow** (postdoctoral researcher)
Feb 2024-Aug 2024	Portland State University (OR, USA) **Postdoctoral researcher** in the group of Prof. Jay Gopalakrishnan
Apr 2021-Jan 2024	TU Wien (Austria) **Postdoctoral university assistant & researcher** in the group of Prof. Joachim Schöberl
Nov 2017-Mar 2021	TU Wien (Austria) **Predoctoral university assistant & researcher** in the group of Prof. Joachim Schöberl

Awards

2022	Promotio sub auspiciis Praesidentis rei publicae (9000€) Awarded by the Austrian president Prof. Alexander Van der Bellen
2018	Diploma thesis award of city Vienna (750€)

Grants

2024	FWF Erwin Schrödinger Fellowship 10.55776/J4824 (PI, 150.362€)

Publications

- **Peer-reviewed**

[1] S. Bartels, A. Bonito, P. Hornung and M. Neunteufel, *Babŭska's paradox in a nonlinear bending-folding model*, Interfaces Free Bound. (2026), published online first. `10.4171/IFB/566`.

[2] J. Li, M. Neunteufel, and L. Zhu, *A novel finite element method for simulating surface plasmon polaritons on complex graphene sheets*, J. Comput. Phys. 542 (2025) 114372. `10.1016/j.jcp.2025.114372`.

[3] J. Gopalakrishnan and M. Neunteufel, *Guided modes of helical waveguides*, Wave Motion 139 (2025) 103621. `10.1016/j.wavemoti.2025.103621`.

[4] A. Sky, M. Neunteufel, J.S. Hale and A. Zilian, *Formulae and transformations for simplicial tensorial finite elements via polytopal templates*, Comput. Math. Appl. 195 (2025) 322–348. `10.1016/j.camwa.2025.07.028`.

[5] A. Dziubek, K. Hu, M. Karow and M. Neunteufel, *Intrinsic mixed finite element methods for linear Cosserat elasticity*, SIAM Journal on Numerical Analysis 63 (2025) 1833–1860. `10.1137/24M1706578`.

[6] G. Fu, M. Neunteufel, J. Schöberl and A. Zdunek, *A four-field mixed formulation for incompressible finite elasticity*, Comput. Methods Appl. Mech. Engrg. 444 (2025) 118082. `10.1016/j.cma.2025.118082`.

[7] A. Pechstein and M. Neunteufel, *Direct coupling of continuum and shell elements in large deformation problems*, Comput. Methods Appl. Mech. Engrg. 442 (2025) 118002. `10.1016/j.cma.2025.118002`.

[8] E.S. Gawlik and M. Neunteufel, *Finite element approximation of the Einstein tensor*, IMA Numer. Anal. (2025). `10.1093/imanum/draf004`.

[9] A. Sky, M. Neunteufel, P. Lewintan, P. Gourgiotis, A. Zilian and P. Neff, *Novel* H^{dev}(Curl)*-conforming elements on regular triangulations and Clough–Tocher splits for the planar relaxed micromorphic model*, Comput. Mech. (2025) `10.1007/s00466-025-02609-1`.

[10] J. Gopalakrishnan, M. Neunteufel, J. Schöberl and M. Wardetzky, *On the improved convergence of lifted distributional Gauss curvature from Regge elements*, Results Appl. Math. 32 (2024) 100511. `10.1016/j.rinam.2024.100511`.

[11] E.S. Gawlik and M. Neunteufel, *Finite element approximation of scalar curvature in arbitrary dimension*, Math. Comp. (2024) `10.1090/mcom/4038`.

[12] M. Neunteufel and J. Schöberl, *The Hellan-Herrmann-Johnson and TDNNS methods for linear and nonlinear shells*, Comput. Struct. 305 (2024) 107543. `10.1016/j.compstruc.2024.107543`.

[13] A. Sky, M. Neunteufel, P. Lewintan, A. Zilian, P. Neff, *Novel H(symCurl)-conforming finite elements for the relaxed micromorphic sequence*, Comput. Methods Appl. Mech. Engrg. 418, Part A (2024) 116494. `10.1016/j.cma.2023.116494`.

[14] J. Gopalakrishnan, M. Neunteufel, J. Schöberl and M. Wardetzky, *Analysis of curvature approximations via covariant curl and incompatibility for Regge metrics*, SMAI J. Comput. Math. 9 (2023) 151–195. `10.5802/smai-jcm.98`.

[15] A. Sky, M. Neunteufel, J. Hale and A. Zilian, *A Reissner-Mindlin plate formulation using symmetric Hu-Zhang elements via polytopal transformations*, Comput. Methods Appl. Mech. Engrg. 416 (2023) 116291. `10.1016/j.cma.2023.116291`.

[16] K. Riehl, M. Neunteufel and M. Hemberg, *Hierarchical confusion matrix for classification performance evaluation* J. R. Stat. Soc. Ser. C Appl. Stat. (2023). `10.1093/jrsssc/qlad057`.

[17] M. Neunteufel, K. Sturm and J. Schöberl, *Numerical shape optimization of the Canham-Helfrich-Evans bending energy.* J. Comput. Phys. 488 (2023) 112218. `10.1016/j.jcp.2023.112218`.

[18] A. Zdunek, M. Neunteufel and W. Rachowicz, *On pressure robustness and independent determination of displacement and pressure in incompressible linear elasticity.* Comput. Methods Appl. Mech. Engrg. 409 (2023) 115714. `10.1016/j.cma.2022.115714`.

[19] A. Sky, M. Neunteufel, I. Münch, J. Schöberl and P. Neff, *Primal and mixed finite element formulations for the relaxed micromorphic model*, Comput. Methods Appl. Mech. Engrg. 399 (2022) 115298. `10.1016/j.cma.2022.115298`.

[20] M. Neunteufel, A. Pechstein and J. Schöberl, *Three-field mixed finite element methods for nonlinear elasticity*, Comput. Methods Appl. Mech. Engrg. 382 (2021) 113857. `10.1016/j.cma.2021.113857`.

[21] A. Sky, M. Neunteufel, I. Münch, J. Schöberl and P. Neff, *A hybrid $H^1 \times H(\mathrm{curl})$ finite element formulation for a relaxed micromorphic continuum model of antiplane shear*, Comput. Mech. 68 (1) (2021) 1–24. `10.1007/s00466-021-02002-8`.

[22] P. Gangl, K. Sturm, M. Neunteufel and J. Schöberl, *Fully and semi-automated shape differentiation in NGSolve*, Struct. Multidiscip. Optim. 63 (3) (2021) 1579–1607. `10.1007/s00158-020-02742-w`.

[23] D. Melching, M. Neunteufel, J. Schöberl and U. Stefanelli, *A finite-strain model for incomplete damage in elastoplastic materials*, Comput. Methods Appl. Mech. Engrg. 374 (2021) 113571. `10.1016/j.cma.2020.113571`.

[24] M. Neunteufel and J. Schöberl, *Avoiding membrane locking with Regge interpolation*, Comput. Methods Appl. Mech. Engrg. 373 (2021) 113524. `10.1016/j.cma.2020.113524`.

[25] M. Neunteufel and J. Schöberl, *Fluid-structure interaction with H(div)-conforming finite elements*, Comput. Struct. 243 (2021) 106402. `10.1016/j.compstruc.2020.106402`.

[26] M. Neunteufel and J. Schöberl, *The Hellan–Herrmann–Johnson method for nonlinear shells*, Comput. Struct. 225 (2019) 106109. `10.1016/j.compstruc.2019.106109`.

- **Preprints**

[1] J. Gopalakrishnan, M. Neunteufel, J. Schöberl and M. Wardetzky, *Generalizing Riemann curvature to Regge metrics* (2023). `arxiv:2311.01603`.

- **Proceedings**

[1] A. Sky, M. Neunteufel, Jack S. Hale and A. Zilian, *Eine schubversteifungsfreie Reissner-Mindlin-Plattenformulierung mittels Hu-Zhang-Element* 15. Fachtagung Baustatik-Baupraxis (2024). https://hdl.handle.net/10993/60522.

[2] T. Danczul, W. Hetebrij, F. Khalighi, L. Kogler, D. Lahaye, E. Luckins, W. Munters, M. Neunteufel, J. Pammer and C. Vuik, *Modeling airflow-driven water droplet removal from a flat surface*, SWI 2020: 157th European Study Group with Industry (2020) 177–214. http://ta.twi.tudelft.nl/users/vuik/papers/SWI22.pdf.

- **Other**

[1] J. Gopalakrishnan, M. Neunteufel, J. Schöberl and M. Wardetzky, *Analysis of curvature approximations via covariant curl and incompatibility for Regge metrics* Oberwolfach report www.mfo.de/occasion/2225/www_view (2022).

[2] M. Neunteufel, A. Pechstein and J. Schöberl, *Mixed Finite Element Methods for Nonlinear Elasticity and Shells* Oberwolfach report www.mfo.de/occasion/2102/www_view (2021).

Teaching experience

SS 2026	Lecture with exercise class *Finite Elements for Differential Geometry*, TU Wien
Spring 2025	Seminar *Continuum mechanics*, Portland State University
SS 2023	Seminar *An introduction to the finite element software NGSolve*, University Luxembourg
SS 2023	Exercise class *Mathematics 2 for Electrical Engineering*, TU Wien
WS 2022/23	Lecture with exercise class *Finite Elements for Differential Geometry*, TU Wien
SS 2022	Exercise class *Numerics of partial differential equations: instationary problems*, TU Wien
WS 2021/22	Exercise class *Numerics of partial differential equations: stationary problems*, TU Wien
SS 2021	Exercise class *Numerics of partial differential equations: instationary problems*, TU Wien
WS 2020/21	Exercise class *Numerics of partial differential equations: stationary problems*, TU Wien
SS 2020	Exercise class *Calculus 2*, TU Wien
WS 2019/20	Exercise class *Calculus 1*, TU Wien
SS 2019	Exercise class *Numerics of partial differential equations: instationary problems*, TU Wien

Mentoring

- **Diploma**

2022	Co-advisor: Edoardo Bonetti *Numerical method for weak gravitational formulation* 10.34726/hss.2022.106702

Conferences & Workshops

- **Invited & contributed talks**

May 2026	ESI: Differential Complexes: Theory, Discretization, and Application, Vienna, Austria
May 2026	20th Austrian Numerical Analysis Day, Graz, Austria
Mar 2026	96th GAMM Annual Meeting, Stuttgart, Germany
Sep 2025	ENUMATH 2025, Heidelberg, Germany
Jul 2025	Workshop: Geometric Mechanics, Structure Preserving Discretization, and Discrete Differential Geometry, Utica (NY), USA
Jul 2025	ICOSAHOM 2025, Montreal, Canada
Jul 2024	SIAM Annual Meeting (AN24), Spokane (WA), USA
Jun 2024	5th NGSolve User Meeting, Vienna, Austria
Apr 2024	8th Cascade RAIN Mathematics Meeting, Portland (OR), USA
Nov 2023	Workshop “Vector- and Tensor-Valued Surface PDEs”, Dresden, Germany
Sep 2023	10th GACM Colloquium on Computational Mechanics, Vienna, Austria
Sep 2023	ENUMATH 2023, Lisbon, Portugal
Jul 2023	Workshop: Geometric Mechanics and Structure Preserving Discretizations of Shell Elasticity, Utica (NY), USA
Jul 2023	4th NGSolve User Meeting, Portland (OR), USA
Jun 2023	1st Jena-Augsburg Meeting on Numerical Analysis, Augsburg, Germany
Apr 2023	17th Austrian Numerical Analysis Day, Vienna, Austria
Mar 2023	PDE Afternoon, TU Wien/University of Vienna/IST Austria, Vienna, Austria
Jul 2022	WAND2022, Salzburg, Austria
Jun 2022	Hilbert Complexis: Analysis, Applications, and Discretizations, Oberwolfach, Germany
May 2022	16th Austrian Numerical Analysis Day, Linz, Austria
Sep 2021	DMV-ÖMG Jahrestagung 2021, Passau, Germany (online)
Jul 2021	ICOSAHOM 2020, Vienna, Austria (online)
Jan 2021	Nonstandard Finite Element Methods, Oberwolfach, Germany (online)
Jul 2019	International Congress on Industrial and Applied Mathematics, Valencia, Spain
Jul 2019	3rd NGSolve User Meeting, Vienna, Austria
Jan 2019	90th GAMM Annual Meeting, Vienna, Austria
Jul 2018	2nd NGSolve User Meeting, Göttingen, Germany
Jun 2018	FEEC and High Order Methods, Oslo, Norway
May 2018	14th Austrian Numerical Analysis Day, Klagenfurt, Austria
Sep 2017	30th Chemnitz FEM Symposium, Strobl, Austria
Jun 2017	1st NGSolve User Meeting, Vienna, Austria
May 2016	12th Austrian Numerical Analysis Day, Innsbruck, Austria

- **Participated**

Sep 2025	50th Anniversary of CAMWA, Heidelberg, Germany
Apr 2023	2nd SFB International Workshop, Vienna, Austria (with poster)
Jan 2020	SWI 2020 Study Group Mathematics with Industry, Fontys Tilburg, Netherlands
Sep 2019	VSM Summer School, Weissensee, Austria
Aug 2019	DK Summer School, Weissensee, Austria
Jan 2019	DK Winter School, Reichenau, Austria
Jan 2018	DK Winter School, Reichenau, Austria
May 2017	13th Austrian Numerical Analysis Day, Salzburg, Austria
Oct 2016	Special Semester on Computational Methods in Science and Engineering, Linz, Austria

Research visits

Oct 2025	University of Minnesota Duluth, USA, at Dr. Li Zhu (with invited talk)
Nov 2024	Santa Clara University, USA, at Prof. Evan Gawlik (with invited talk)
Jul 2023	Suny Polytechnic Institute, USA, at Prof. Andrea Dziubek (with seminar course)
Jul 2023	Portland State University, USA, at Prof. Jay Gopalakrishnan
Mai 2023	University Luxembourg, Luxembourg, at Prof. Andreas Zilian (with invited talk)
Feb 2023	University of Hawai’i at Manoa, USA, at Prof. Evan Gawlik (with invited talk and seminar)
Jan 2020	University of Stuttgart, Germany, at Prof. Manfred Bischoff (with invited talk)
Oct 2019	Johannes Kepler University Linz, Austria, at Prof. Astrid Pechstein (with invited talk)
Jan 2018	University of Göttingen, Germany, at Prof. Christoph Lehrenfeld (with invited talk)

Scientific training

Oct 2023 Course *FWF ESPRIT & Schrödinger - The missing manual* (Austria)
Apr 2022 Seminar *Essentials in National Research Funding* (Austria)
Jun 2020 Course in proposal writing
Jan 2019 Course in scientific writing
Jan 2018 Course in presentation techniques

Organization of scientific events

Jul 2025 Co-organizer of workshop *Geometric Mechanics, Structure Preserving Discretization, and Discrete Differential Geometry*, Utica (NY), USA
Sep 2023 Member organizing committee of *10th GACM Colloquium on Computational Mechanics*, Vienna

- **Technical assistant**

Jul 2021 ICOSAHOM 2020, Vienna
Jul 2019 3rd NGSolve User-Meeting, Vienna
Feb 2019 90th GAMM Annual Meeting, Vienna

Journal referee

- **Member**

Sep 2024-present Editorial board Computers and Mathematics with Applications (CAMWA)

- **Journals (alphabetic order)**
  Advances in Computational Mathematics; Applied Mathematics & Optimization; Applied Mathematical Modeling; Computational Mechanics; Computers and Mathematics with Applications; Computers & Structures; Engineering Analysis with Boundary Elements; Foundations of Computational Mathematics; IMA Journal of Numerical Analysis; Journal of Computational and Applied Mathematics; Journal of Intelligent Material Systems and Structures; Journal of Numerical Mathematics; Journal of Scientific Computing; Mathematics of Computation; SIAM Journal of Numerical Analysis; SIAM Journal on Scientific Computing

Co-authors (alphabetic order)

Sören Bartels (University of Freiburg)
Andrea Dziubek (SUNY Polytechnic Institute)
Guosheng Fu (University of Notre Dame)
Peter Gangl (JKU Linz)
Evan Gawlik (Santa Clara University)
Jay Gopalakrishnan (Portland State University)
Panos Gourgiotis (National Technical University of Athens)
Jack Hale (University Luxembourg)
Martin Hemberg (Harvard University)
Peter Hornung (TU Dresden)
Kaibo Hu (University of Edinburgh)
Michael Karow (TU Berlin)
Peter Lewintan (University Duisburg-Essen)
Jichun Li (University of Nevada Las Vegas)
David Melching (German Aerospace Center)
Ingo Münch (Technical University Dortmund)
Patrizio Neff (University Duisburg-Essen)
Astrid Pechstein (JKU Linz)
Waldemar Rachowicz (Cracow University of Technology)
Kevin Riehl (ETH Zurich)
Joachim Schöberl (TU Wien)
Adam Sky (University Luxembourg)
Ulisse Stefanelli (University of Vienna)
Kevin Sturm (TU Wien)
Max Wardetzky (University of Göttingen)
Adam Zdunek (HB BerRit)
Andreas Zilian (University Luxembourg)
Li Zhu (University of Minnesota Duluth)

Research experience

2024-present	FWF Erwin Schrödinger Fellowship 10.55776/J4824 (PI)
2021-2023	SFB Taming Complexity in Partial Differential Systems, Automated Discretization in Multiphysics, Austrian Science Fund (FWF) 1429331/F 6511-N36 (member)
Feb-Oct 2021	Multiface Observation of Interface Evolution, AC2T research GmbH 1931712 (member)
2017-2021	Completed "Dissipation and Dispersion in Nonlinear PDEs" (nPDE) DK program (member)
2019-2021	Completed "Vienna School of Mathematics" (member)
2017-2019	Numerics of Dissipation, Austrian Science Fund (FWF) W 1245 (member)

Programming & Software skills

Languages	C, C++, Java, Python
Software	Matlab, Maple, NGSolve, NGSDiffGeo

Languages

German (first language)
English (fluent)
Italian (basic)